

Операционные системы

Командная оболочка Midnight Commander

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Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

1. Цели и задачи работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

- 1 Изучить возможности Midnight Commander
- 2 Изучить редактор Midnight Commander

2. Процесс выполнения лабораторной работы

Работа с Midnight Commander

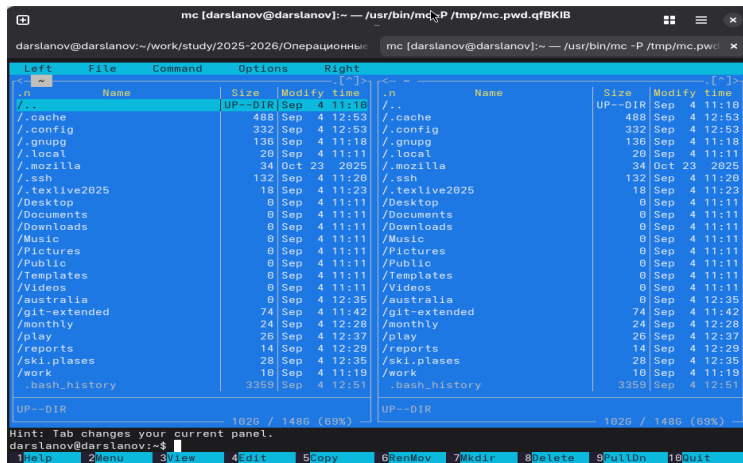


Рисунок 1: Запуск mc

Работа с Midnight Commander

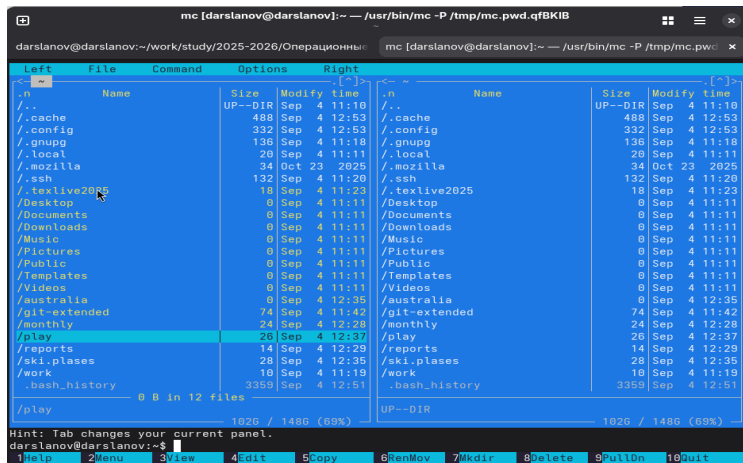


Рисунок 2: Выделение

Работа с Midnight Commander

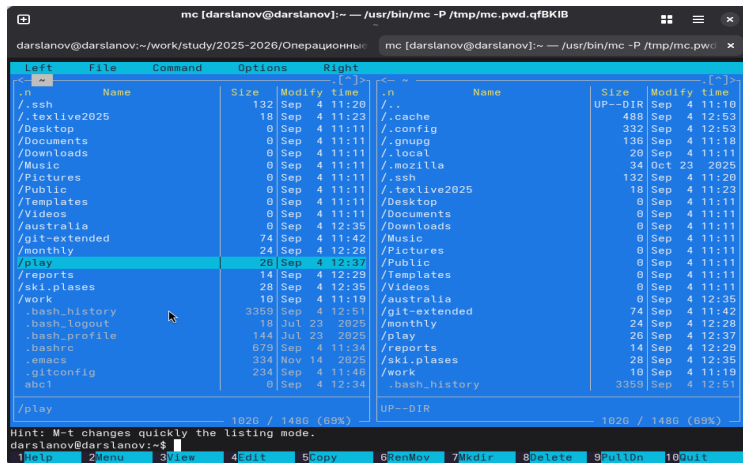


Рисунок 3: Отмена

Работа с Midnight Commander

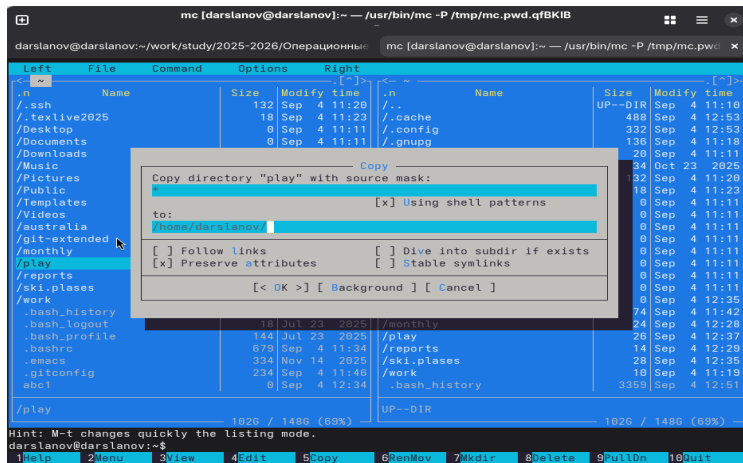


Рисунок 4: Копирование

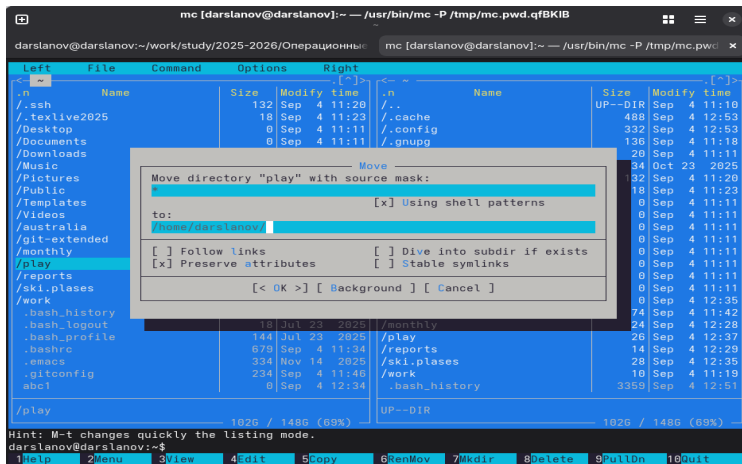


Рисунок 5: Перемещение

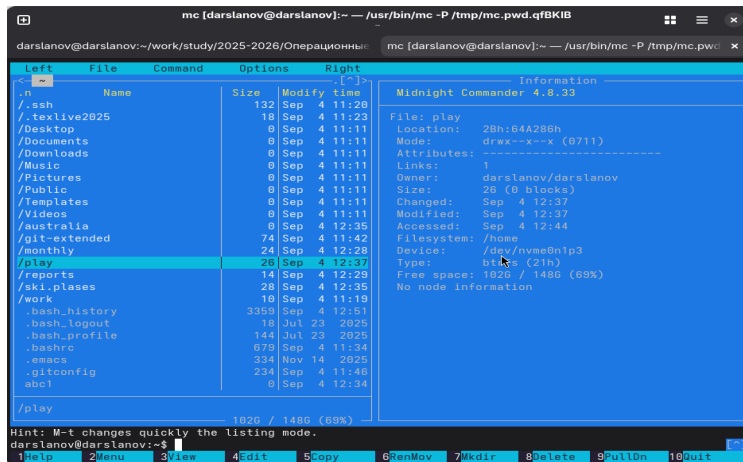


Рисунок 6: Информация

Работа с Midnight Commander

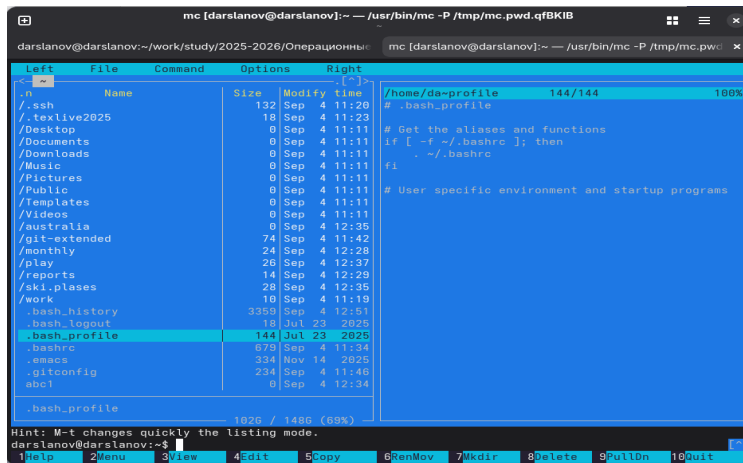


Рисунок 7: Быстрый просмотр

Работа с Midnight Commander

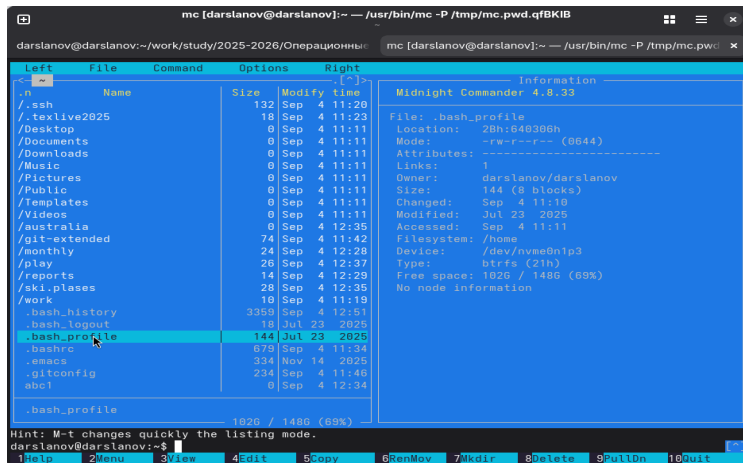


Рисунок 8: Информация

Работа с Midnight Commander

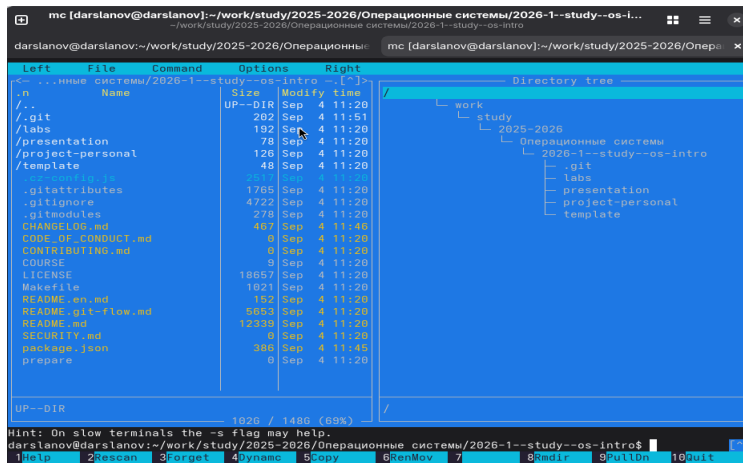
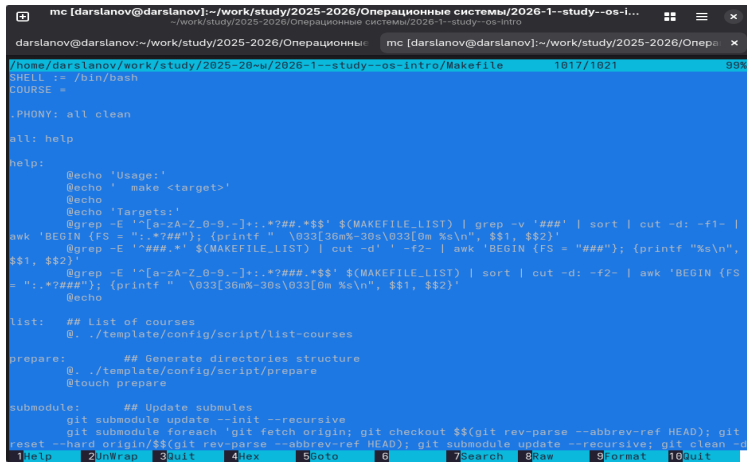


Рисунок 9: Дерево каталогов



```
mc [darсланов@darсланов]:~/work/study/2025-2026/Операционные системы/2026-1--study--os-i...
~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro
darсланов@darсланов:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro
/home/darсланов/work/study/2025-2026-20-м/2026-1--study--os-intro/Makefile 1017/1021 99%
SHELL := /bin/bash
COURSE =

.PHONY: all clean

all: help

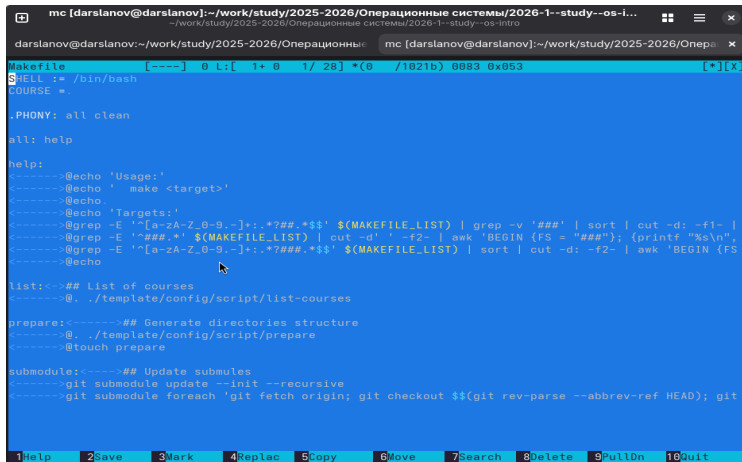
help:
    @echo 'Usage:'
    @echo '  make <target>'
    @echo
    @echo 'Targets:'
    @grep -E '^[a-zA-Z_0-9-]+:.*?##.*$$' $(MAKEFILE_LIST) | grep -v '###' | sort | cut -d: -f1- |
    awk 'BEGIN {FS = ":.*?##"}; {printf " \033[36m%-30s\033[0m %s\n", $$1, $$2}'
    @grep -E '^[^##].*' $(MAKEFILE_LIST) | cut -d' ' -f2- | awk 'BEGIN {FS = "###"}; {printf "%s\n",
    $$1, $$2}'
    @grep -E '^[a-zA-Z_0-9-]+:.*?##.*$$' $(MAKEFILE_LIST) | sort | cut -d: -f2- | awk 'BEGIN {FS
    = ":.*?##"}; {printf " \033[36m%-30s\033[0m %s\n", $$1, $$2}'
    @echo

list:    ## List of courses
    @. ./template/config/script/list-courses

prepare:    ## Generate directories structure
    @. ./template/config/script/prepare
    @touch prepare

submodule:    ## Update submules
    git submodule update --init --recursive
    git submodule foreach 'git fetch origin; git checkout $$ (git rev-parse --abbrev-ref HEAD); git
    reset --hard origin/$(git rev-parse --abbrev-ref HEAD); git submodule update --recursive; git clean -d
```

Рисунок 10: Просмотр содержимого текстового файла



```
mc [darсланов@darсланов]:~/work/study/2025-2026/Операционные системы/2026-1--study--os-i...
~/work/study/2025-2026/Операционные системы/2026-1--study--os-Intro
darсланов@darсланов:~/work/study/2025-2026/Операционные системы/2026-1--study--os-Intro
mc [darсланов@darсланов]:~/work/study/2025-2026/Операционные системы/2026-1--study--os-Intro

Makefile [----] 0 L: [ 1+ 0 1/ 28] *(0 /1021b) 0083 0x053 [*][X]
SHELL := /bin/bash
COURSE =.

.PHONY: all clean

all: help

help:
<----->@echo 'Usage:'
<----->@echo '  make <target>'
<----->@echo
<----->@echo 'Targets:'
<----->@grep -E '^[a-zA-Z_0-9-]+:.*?###.*$$' $(MAKEFILE_LIST) | grep -v '###' | sort | cut -d: -f1- |
<----->@grep -E '^[^###.*] $(MAKEFILE_LIST) | cut -d' ' -f2- | awk 'BEGIN {FS = "###"}; {printf "%s\n",
<----->@grep -E '^[a-zA-Z_0-9-]+:.*?###.*$$' $(MAKEFILE_LIST) | sort | cut -d: -f2- | awk 'BEGIN {FS
<----->@echo

list:<-->## List of courses
<----->@. ./template/config/script/list-courses

prepare:<----->## Generate directories structure
<----->@. ./template/config/script/prepare
<----->@touch prepare

submodule:<----->## Update submules
<----->git submodule update --init --recursive
<----->git submodule foreach 'git fetch origin; git checkout $$ (git rev-parse --abbrev-ref HEAD); git
```

Рисунок 11: Отредактируем содержимое текстового файла без сохранения результатов

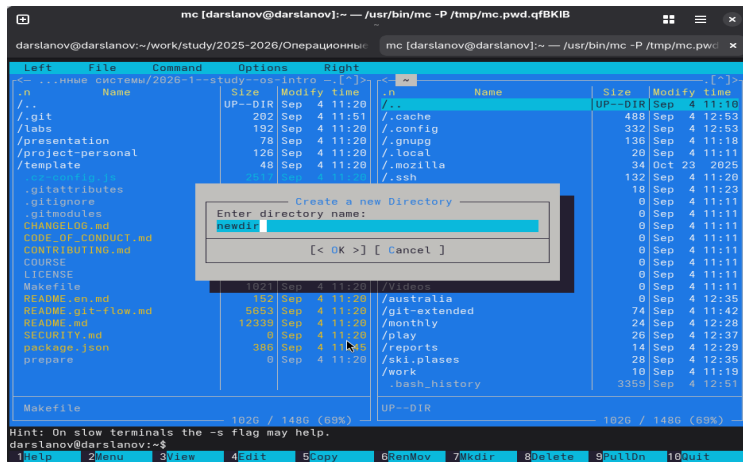


Рисунок 12: Создание каталога

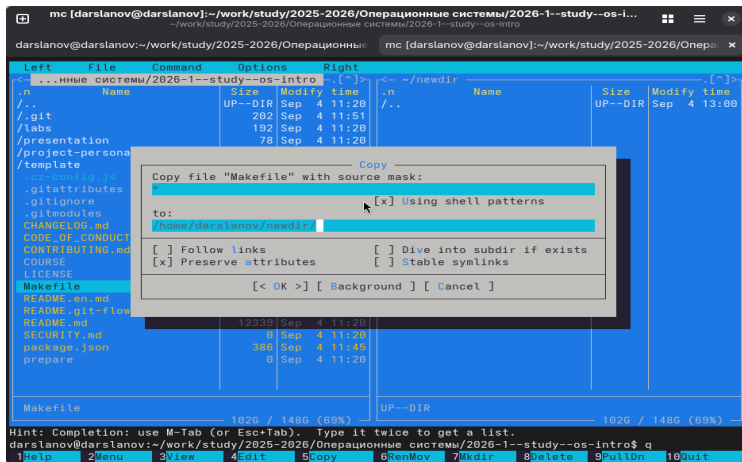


Рисунок 13: Копирование в файлов в созданный каталог

Работа с Midnight Commander

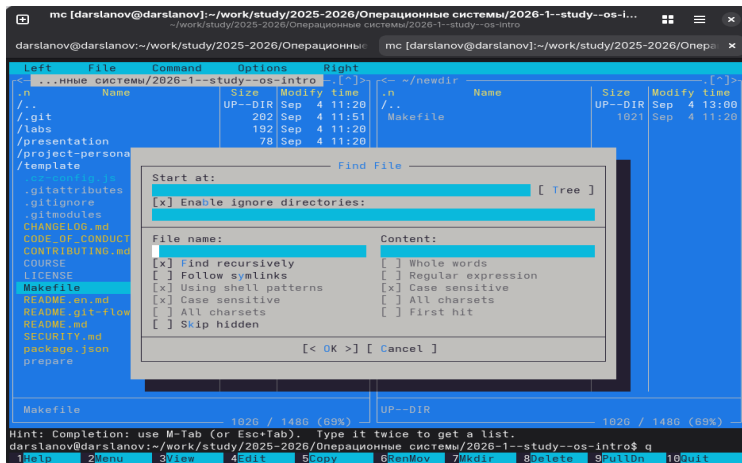


Рисунок 14: Поиск файлов

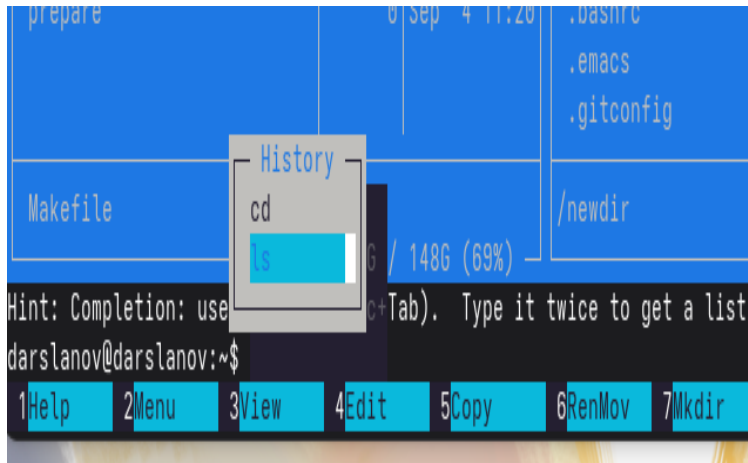
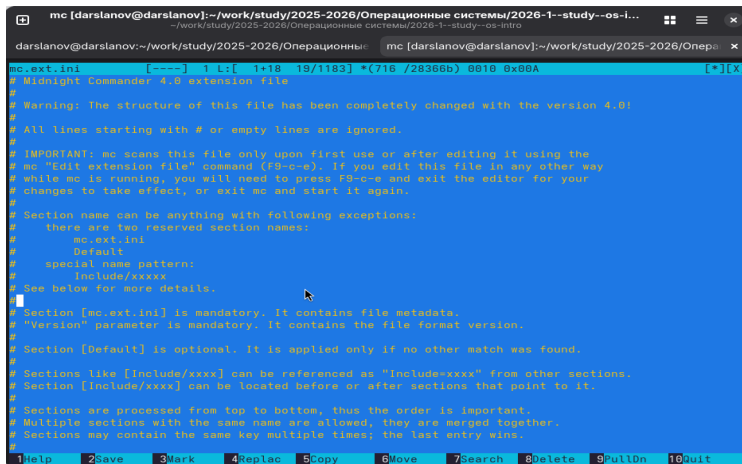


Рисунок 15: История команд



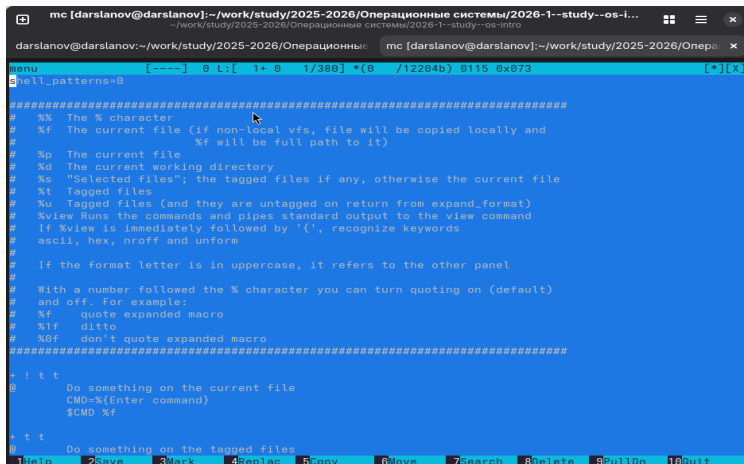
Рисунок 16: Переход в домашний каталог

The image shows a screenshot of the Midnight Commander (MC) file manager interface. The title bar at the top indicates the current directory is ~/work/study/2025-2026/Операционные системы/2026-1--study--os-i... and the active window is mc [darсланov@darсланov]:~/work/study/2025-2026/Операционные системы/2026-1--study--os-i... The main content area displays the contents of the mc.ext.ini file. The text is as follows:

```
mc.ext.ini [----] 1 L: [ 1+18 19/1183] *(716 /28366b) 0010 0x00A [*][X]
# Midnight Commander 4.8 extension file
#
# Warning: The structure of this file has been completely changed with the version 4.0!
#
# All lines starting with # or empty lines are ignored.
#
# IMPORTANT: mc scans this file only upon first use or after editing it using the
# mc "Edit extension file" command (F9-c-e). If you edit this file in any other way
# while mc is running, you will need to press F9-c-e and exit the editor for your
# changes to take effect, or exit mc and start it again.
#
# Section name can be anything with following exceptions:
#   there are two reserved section names:
#       mc.ext.ini
#       Default
#   special name pattern:
#       Include/xxxx
# See below for more details.
#
# Section [mc.ext.ini] is mandatory. It contains file metadata.
# "Version" parameter is mandatory. It contains the file format version.
#
# Section [Default] is optional. It is applied only if no other match was found.
#
# Sections like [Include/xxxx] can be referenced as "Include=xxxx" from other sections.
# Section [Include/xxxx] can be located before or after sections that point to it.
#
# Sections are processed from top to bottom, thus the order is important.
# Multiple sections with the same name are allowed, they are merged together.
# Sections may contain the same key multiple times; the last entry wins.
```

At the bottom of the window, there is a status bar with the following shortcuts: 1Help, 2Save, 3Mark, 4Replac, 5Copy, 6Move, 7Search, 8Delete, 9PullDn, 10Quit.

Рисунок 17: Просмотр файла расширений

A screenshot of the Midnight Commander (MC) application window. The window title is "mc [darslanov@darslanov]:~/work/study/2025-2026/Операционные системы/2026-1--study--os-i...". The address bar shows the current directory: "darslanov@darslanov:~/work/study/2025-2026/Операционные". The main pane displays the file menu, which is a list of options for manipulating files and directories. The menu items are: "menu", "Shell_patterns=0", a separator line, a list of format letters and their descriptions (%, %f, %p, %d, %s, %t, %u, %view, %ascii, %hex, %nroff, %uniform), a list of actions (!, @, +, t, t), and a bottom bar with numbered shortcuts (1-10) for Help, Save, Mark, Replace, Copy, Move, Search, Delete, PullDn, and Quit. The background is blue, and the text is white and yellow.

```
mc [darslanov@darslanov]:~/work/study/2025-2026/Операционные системы/2026-1--study--os-i...
~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro
darslanov@darslanov:~/work/study/2025-2026/Операционные mc [darslanov@darslanov]:~/work/study/2025-2026/Онепа

menu [----] 0 L: 1+ 0 1/380] *(0 /12204b) 0115 0x073 [*][X]
Shell_patterns=0

#####
# %% The % character
# %f The current file (if non-local vfs, file will be copied locally and
# %f will be full path to it)
# %p The current file
# %d The current working directory
# %s "Selected files"; the tagged files if any, otherwise the current file
# %t Tagged files
# %u Tagged files (and they are untagged on return from expand_format)
# %view Runs the commands and pipes standard output to the view command
# If %view is immediately followed by '(', recognize keywords
# ascii, hex, nroff and uniform
#
# If the format letter is in uppercase, it refers to the other panel
#
# With a number followed the % character you can turn quoting on (default)
# and off. For example:
# %f quote expanded macro
# %!f ditto
# %0f don't quote expanded macro
#####
+ ! t t
@ Do something on the current file
  CMD=%(Enter command)
  $CMD %f
+ t t
@ Do something on the tagged files
1Help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullDn 10Quit
```

Рисунок 18: Просмотр файла меню

Работа с Midnight Commander

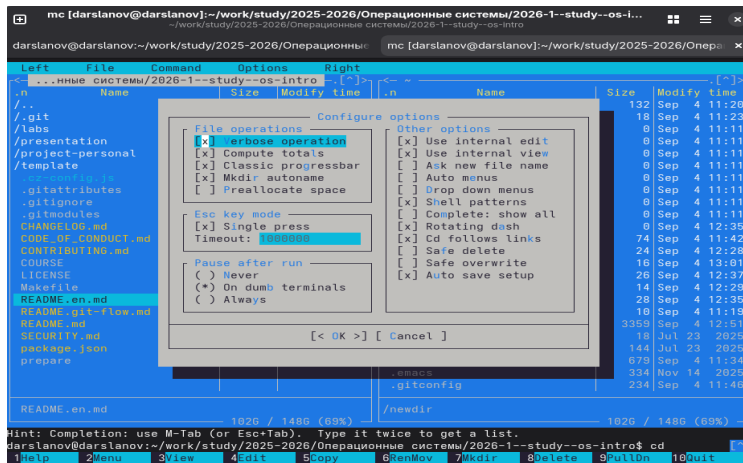


Рисунок 19: Конфигурация

Работа с Midnight Commander

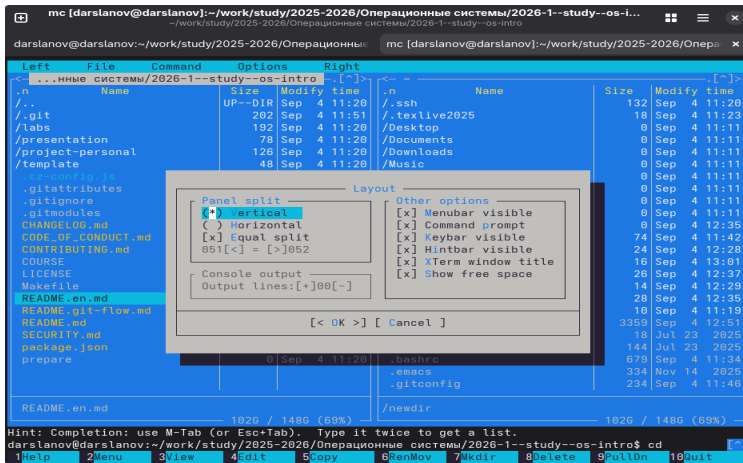


Рисунок 20: Внешний вид

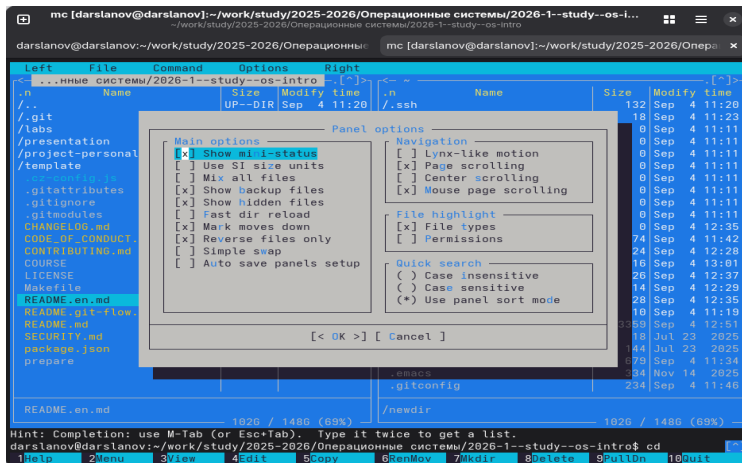


Рисунок 21: Настройки панелей

Работа с Midnight Commander

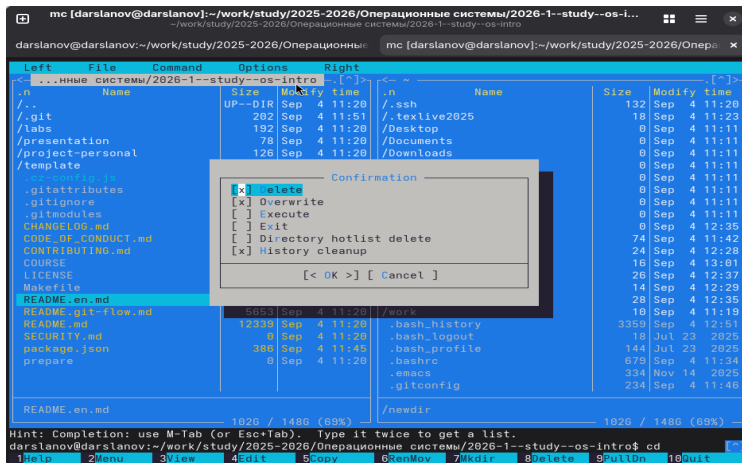


Рисунок 22: Подтверждение

Работа с Midnight Commander

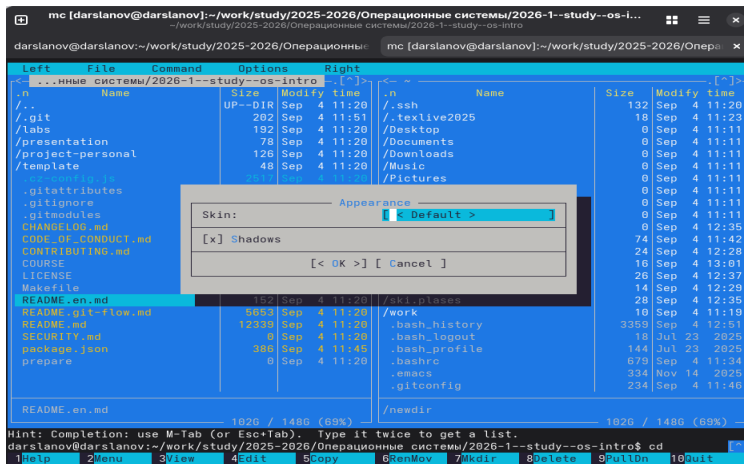


Рисунок 23: Оформление

Работа с Midnight Commander

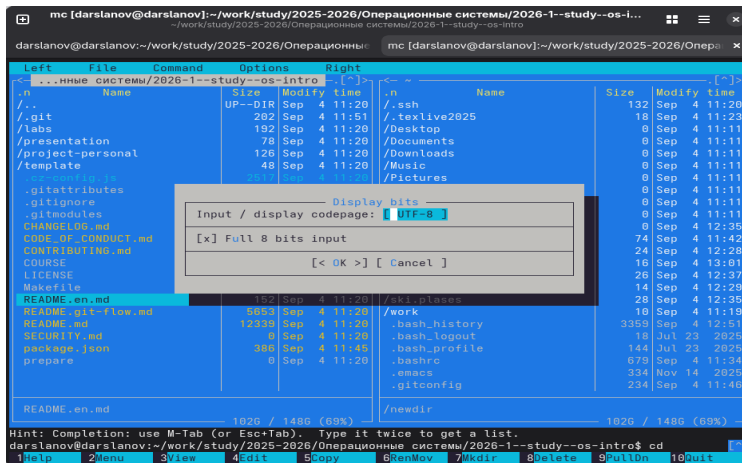


Рисунок 24: Кодировка символов

Работа с Midnight Commander

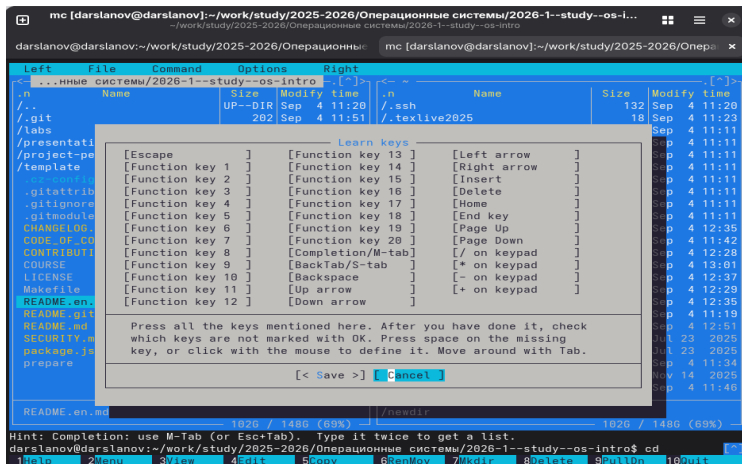
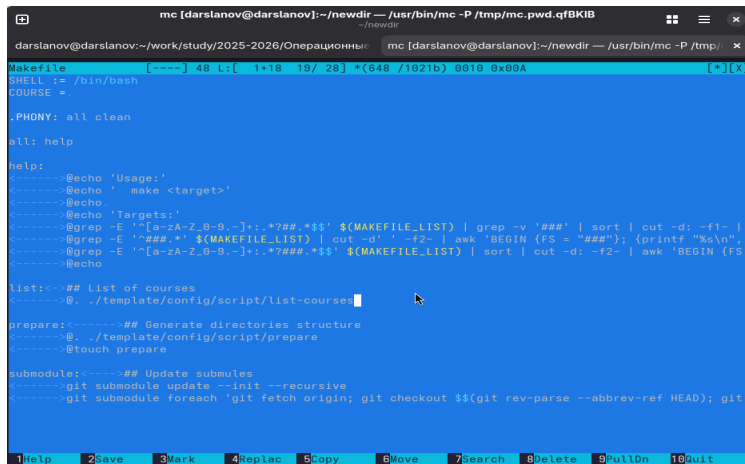


Рисунок 25: Распознавание клавиш

Работа с редактором Midnight Commander



```
mc [darlanov@darlanov]:~/newdir — /usr/bin/mc -P /tmp/mc.pwd.qfBKIB
~ /newdir
darlanov@darlanov:~/work/study/2025-2026/Операционные mc [darlanov@darlanov]:~/newdir — /usr/bin/mc -P /tmp/

Makefile [----] 48 L: [ 1+18 19/ 28 ] *(648 /1021b) 0010 0x00A [.*][X]
SHELL := /bin/bash
COURSE =

.PHONY: all clean

all: help

help:
<----->@echo 'Usage:'
<----->@echo '  make <target>'
<----->@echo
<----->@echo 'Targets:'
<----->@grep -E '^[a-zA-Z_0-9-]+:.*?###.*$$' $(MAKEFILE_LIST) | grep -v '###' | sort | cut -d: -f1- |
<----->@grep -E '^[^###.*] $(MAKEFILE_LIST) | cut -d' ' -f2- | awk 'BEGIN {FS = "###"}; {printf "%s\n",
<----->@grep -E '^[a-zA-Z_0-9-]+:.*?7###.*$$' $(MAKEFILE_LIST) | sort | cut -d: -f2- | awk 'BEGIN {FS
<----->@echo

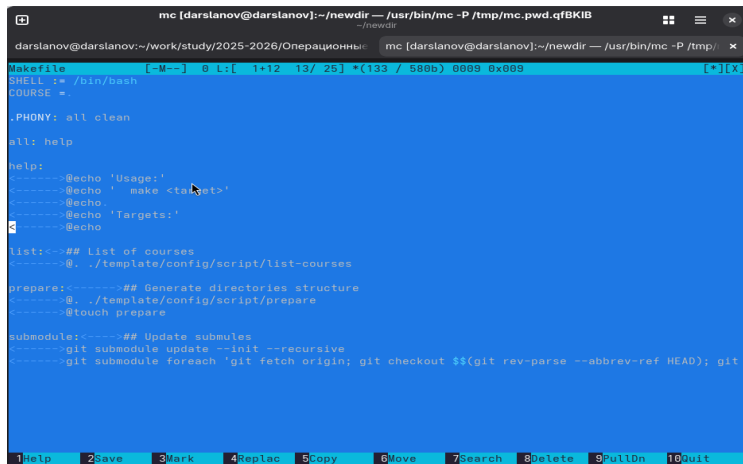
list:<-->## List of courses
<----->@. ./template/config/script/list-courses

prepare:<----->## Generate directories structure
<----->@. ./template/config/script/prepare
<----->@touch prepare

submodule:<----->## Update submules
<----->git submodule update --init --recursive
<----->git submodule foreach 'git fetch origin; git checkout $$ (git rev-parse --abbrev-ref HEAD); git
```

Рисунок 26: Файл с текстом

Работа с редактором Midnight Commander

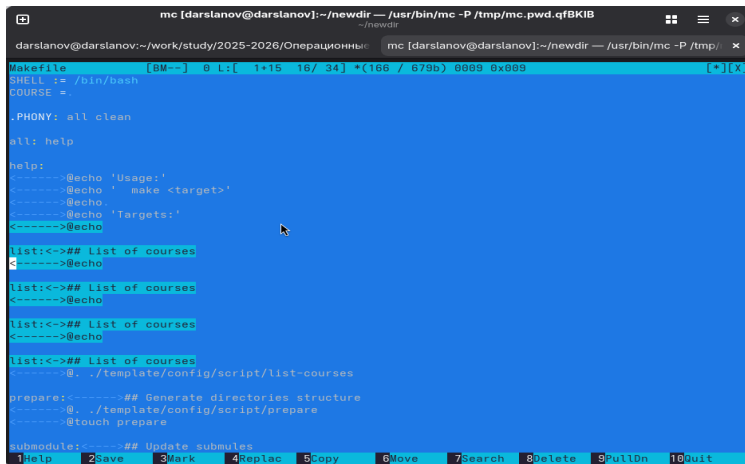


```
mc [darlanov@darlanov]:~/newdir — /usr/bin/mc -P /tmp/mc.pwd.qfBKIB
~/newdir
darlanov@darlanov:~/work/study/2025-2026/Операционные системы — mc [darlanov@darlanov]:~/newdir — /usr/bin/mc -P /tmp/
Makefile [-M--] 0 L: [ 1+12 13/ 25] *(133 / 580b) 0009 0x009 [*][X]
SHELL := /bin/bash
COURSE = .
.PHONY: all clean
all: help
help:
----->@echo 'Usage:'
----->@echo '  make <target>'
----->@echo .
----->@echo 'Targets:'
----->@echo
list:<----->## List of courses
----->@. ./template/config/script/list-courses
prepare:<----->## Generate directories structure
----->@. ./template/config/script/prepare
----->@touch prepare
submodule:<----->## Update submodules
----->git submodule update --init --recursive
----->git submodule foreach 'git fetch origin; git checkout $$ (git rev-parse --abbrev-ref HEAD); git
```

1 Help 2 Save 3 Mark 4 Replac 5 Copy 6 Move 7 Search 8 Delete 9 PullDn 10 Quit

Рисунок 27: Файл с текстом

Работа с редактором Midnight Commander



```
mc [darslanov@darslanov]:~/newdir — /usr/bin/mc -P /tmp/mc.pwd.qfBKIB
~/newdir
darslanov@darslanov:~/work/study/2025-2026/Операционные системы — /usr/bin/mc -P /tmp/

Makefile [BM--] 0 L: [ 1+15 16/ 34] *(166 / 679b) 0009 0x009 [*][X]
SHELL := /bin/bash
COURSE = .

.PHONY: all clean

all: help

help:
<----->@echo 'Usage:'
<----->@echo '  make <target>'
<----->@echo .
<----->@echo 'Targets:'
<----->@echo

list:<->## List of courses
<----->@echo

list:<->## List of courses
<----->@echo

list:<->## List of courses
<----->@echo

list:<->## List of courses
<----->@. ./template/config/script/list-courses

prepare:<----->## Generate directories structure
<----->@. ./template/config/script/prepare
<----->@touch prepare

submodule:<----->## Update submules
1Help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullDn 10Quit
```

Рисунок 28: Копирование фрагмента

Работа с редактором Midnight Commander

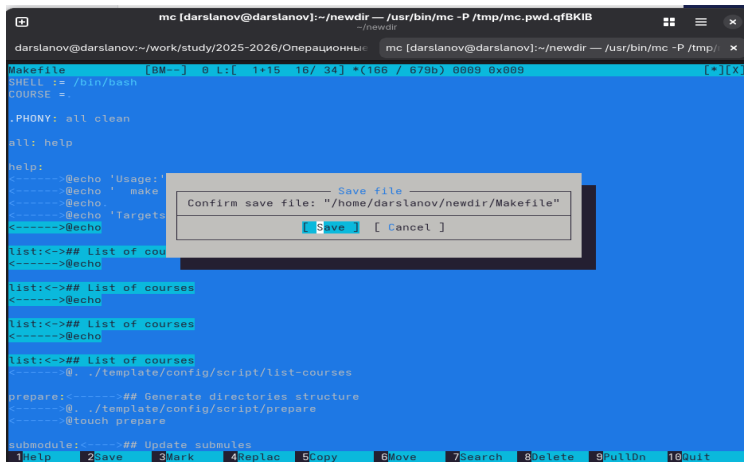
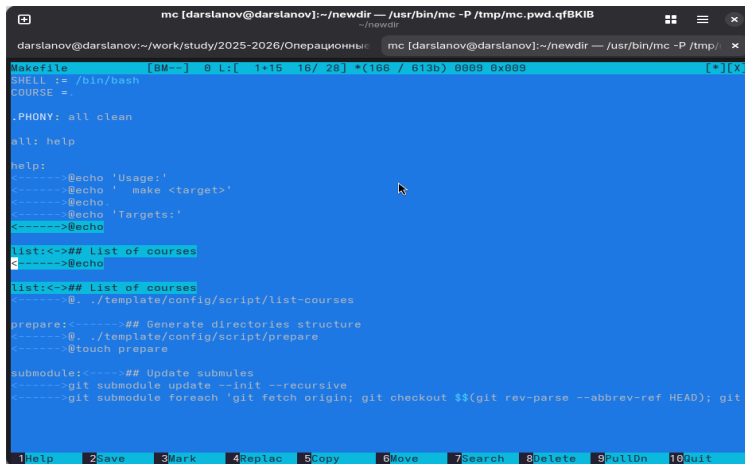


Рисунок 29: Сохранение

Работа с редактором Midnight Commander



The screenshot shows the Midnight Commander (MC) interface. The title bar indicates the user is 'mc [darсланov@darсланov]:~/newdir' and the current directory is '/usr/bin/mc -P /tmp/mc.pwd.qfBKIB'. The main window displays a file named 'Makefile' with a status bar at the bottom showing '1Help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullDn 10Quit'. The content of the 'Makefile' is as follows:

```
Makefile [BM--] 0 L: [ 1+15 16/ 28] *(166 / 613b) 0009 0x009 [*][X]
SHELL := /bin/bash
COURSE = .

.PHONY: all clean

all: help

help:
<----->@echo 'Usage:'
<----->@echo ' make <target>'
<----->@echo .
<----->@echo 'Targets:'
<----->@echo

list:<-->## List of courses
<----->@echo

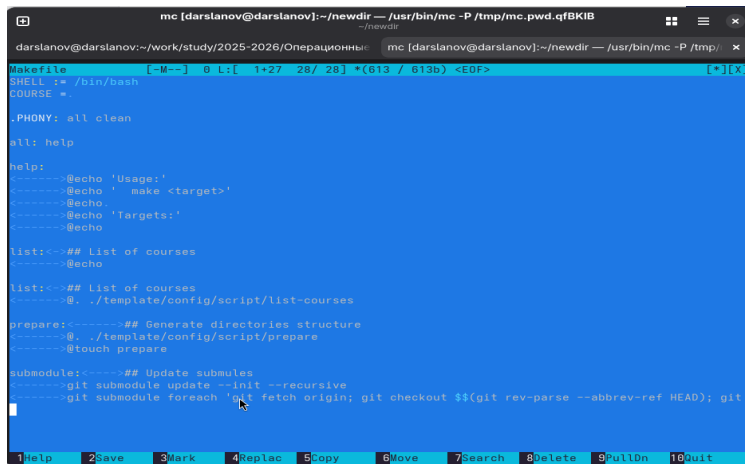
list:<-->## List of courses
<----->@. ./template/config/script/list-courses

prepare:<----->## Generate directories structure
<----->@. ./template/config/script/prepare
<----->@touch prepare

submodule:<----->## Update submules
<----->git submodule update --init --recursive
<----->git submodule foreach 'git fetch origin; git checkout $$ (git rev-parse --abbrev-ref HEAD); git
```

Рисунок 30: Отмена

Работа с редактором Midnight Commander



The screenshot shows the Midnight Commander (MC) interface. The title bar indicates the current directory is `~/newdir` and the file being edited is `Makefile`. The main window displays the contents of the `Makefile` file, which includes comments and commands for setting the shell, course name, and preparing the environment. The bottom status bar shows various navigation and editing commands: 1Help, 2Save, 3Mark, 4Replac, 5Copy, 6Move, 7Search, 8Delete, 9PullDn, 10Quit.

```
mc [darlanov@darlanov]:~/newdir — /usr/bin/mc -P /tmp/mc.pwd.qfBKIB
~/newdir
darlanov@darlanov:~/work/study/2025-2026/Операционные: mc [darlanov@darlanov]:~/newdir — /usr/bin/mc -P /tmp/

Makefile      [-M--]  0 L:[ 1+27 28/ 28] *(613 / 613b) <EOF>      [+][X]
SHELL := /bin/bash
COURSE = .

.PHONY: all clean

all: help

help:
----->@echo 'Usage:'
----->@echo '  make <target>'
----->@echo
----->@echo 'Targets:'
----->@echo

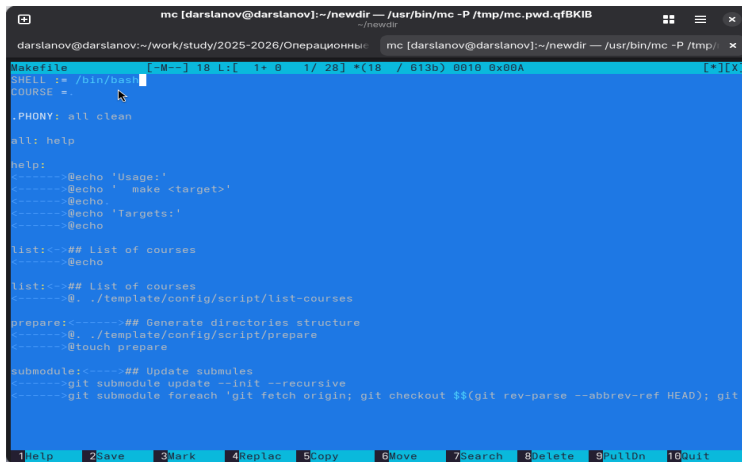
list:<->## List of courses
----->@echo

list:<->## List of courses
----->@. ./template/config/script/list-courses

prepare:<->## Generate directories structure
----->@. ./template/config/script/prepare
----->@touch prepare

submodule:<->## Update submules
----->git submodule update --init --recursive
----->git submodule foreach 'git fetch origin; git checkout $$ (git rev-parse --abbrev-ref HEAD); git
```

Рисунок 31: Переход в конец файла

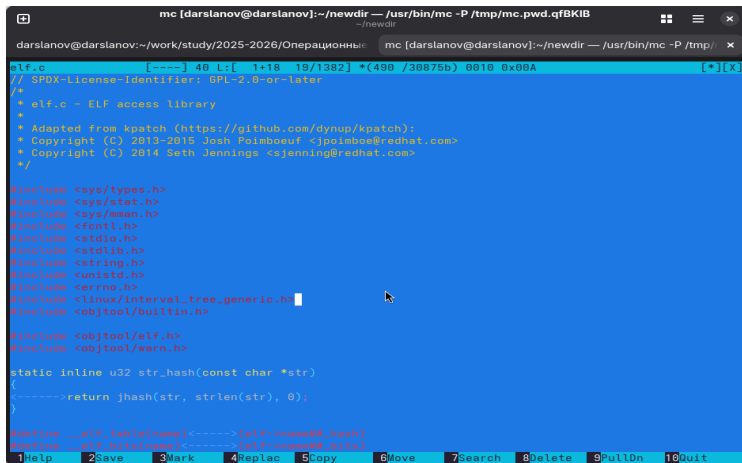


```
mc [darlanov@darlanov]:~/newdir — /usr/bin/mc -P /tmp/mc.pwd.qfBKIB
~/newdir
darlanov@darlanov:~/work/study/2025-2026/Операционные: mc [darlanov@darlanov]:~/newdir — /usr/bin/mc -P /tmp/
Makefile [-M--] 18 L: [ 1+ 8 1/ 28 ] *(18 / 613b) 0010 0x00A [*][X]
SHELL := /bin/bash
COURSE = .
.PHONY: all clean
all: help
help:
<----->@echo 'Usage:'
<----->@echo ' make <target>'
<----->@echo.
<----->@echo 'Targets:'
<----->@echo
list:<-->## List of courses
<----->@echo
list:<-->## List of courses
<----->@. ./template/config/script/list-courses
prepare:<----->## Generate directories structure
<----->@. ./template/config/script/prepare
<----->@touch prepare
submodule:<----->## Update submules
<----->git submodule update --init --recursive
<----->git submodule foreach 'git fetch origin; git checkout $$ (git rev-parse --abbrev-ref HEAD); git
```

1 Help 2 Save 3 Mark 4 Replac 5 Copy 6 Move 7 Search 8 Delete 9 PullDn 10 Quit

Рисунок 32: Переход в начало файла

Работа с редактором Midnight Commander



The screenshot shows the Midnight Commander (MC) file editor interface. The title bar indicates the current directory is `~/newdir` and the file being edited is `elf.c`. The status bar at the bottom shows various keyboard shortcuts: 1Help, 2Save, 3Mark, 4Replac, 5Copy, 6Move, 7Search, 8Delete, 9PullDn, 10Quit. The main editing area displays the source code of `elf.c`, which is an ELF access library. The code includes several system headers and defines a `str_hash` function. The cursor is positioned at the end of the `#include <linux/interval_tree_generic.h>` line.

```
elf.c      [----] 40 L: [ 1+18 19/1382] *(498 /30875b) 0010 0x00A [*][X]
// SPDX-License-Identifier: GPL-2.0-or-later
/*
 * elf.c - ELF access library
 *
 * Adapted from kpatch (https://github.com/dynup/kpatch):
 * Copyright (C) 2013-2015 Josh Poimboeuf <jpoimboe@redhat.com>
 * Copyright (C) 2014 Seth Jennings <sjenning@redhat.com>
 */

#include <sys/types.h>
#include <sys/stat.h>
#include <sys/mman.h>
#include <fcntl.h>
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <errno.h>
#include <linux/interval_tree_generic.h>
#include <objtool/builtin.h>

#include <objtool/elf.h>
#include <objtool/warn.h>

static inline u32 str_hash(const char *str)
{
    ----->return jhash(str, strlen(str), 0);
}

#define __elf_inbin(name) <----->{elf->name##.hash}
#define __elf_bits(name) <----->{elf->name##.bits}
```

Рисунок 33: Файл с программой

Работа с редактором Midnight Commander

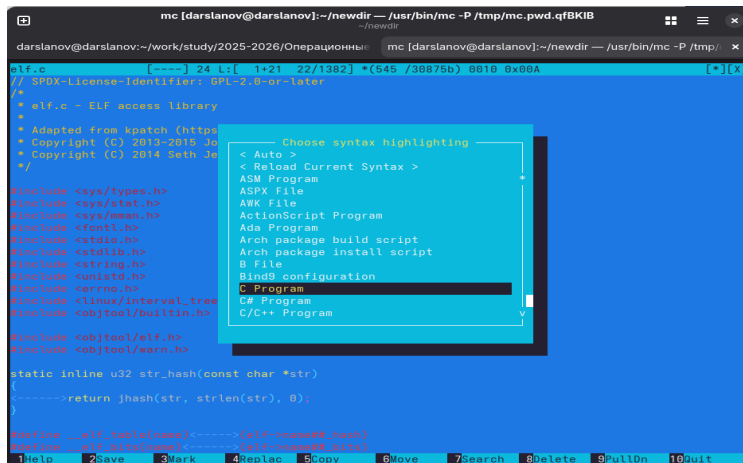


Рисунок 34: Цветовыделение синтаксиса

3. Выводы по проделанной работе

В данной работе мы ознакомились с инструментами поиска файлов и фильтрации текстовых данных. А также приобрели практические навыки по управлению процессами.